The documentation and process conversion measures necessary to comply with this revision shall be completed by 29 July 2005.

INCH-POUND

MIL-PRF-19500/545F
29 April 2005
SUPERSEDING
MIL-PRF-19500/545E
13 March 2003

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, POWER, TYPES 2N5151, 2N5153, 2N5151L, 2N5153L, 2N5151U3, AND 2N5153U3, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, power transistors for use in high-speed power-switching applications. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to TO-205), figures 2, 3, and 4 (JANHC and JANKC), and figure 5 (U3).

* 1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Types	P_T T_A = +25°C (1)	P_T T_C = +25°C (1)	$R_{\theta JA}$ (2)	$R_{\theta JC}$ (2)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_C (3)	Reverse pulse energy (4)	T_{STG} and T_J
	W	W	°C/W	°C/W	V dc	V dc	V dc	A dc	A dc	mj	°C
2N5151, L	1 (5)	11.8 (6)	175	10	100	80	5.5	2	10	15	-65 to
2N5153, L	1 (5)	11.8 (6)	175	10	100	80	5.5	2	10	15	+ 200
2N5151U3	1.16 (7)	100 (8)	150	1.75	100	80	5.5	2	10	15	-65 to
2N5153U3	1.16 (7)	100 (8)	150	1.75	100	80	5.5	2	10	15	+ 200

* (1) See figure 6, 7, 8, and 9 for temperature-power derating curves.
* (2) See figure 10, 11, and 12 for transient thermal impedance graph.
(3) This value applies for Pw ≤ 8.3 ms, duty cycle ≤ 1 percent.
* (4) This rating is based on the capability of the transistors to operate safely in the unclamped inductive load energy test circuit of figure 13.
(5) Derate linearly 5.7 mW/°C for T_A > +25°C.
(6) Derate linearly 66.7 mW/°C for T_C > +25°C.
(7) Derate linearly 6.63 mW/°C for T_A > +25°C.
(8) Derate linearly 571 mW/°C for T_C > +25°C.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A

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1.4 Primary electrical characteristics at T_C = +25°C.

Limits	h_{FE2} (1) V_{CE} = 5 V I_C = 2.5 A dc		$\|h_{fe}\|$ V_{CE} = 5 V I_C = 500 mA dc f = 10 MHz		$V_{BE(sat)2}$ (1) I_C = 5 A dc I_B = 500 mA dc	$V_{CE(sat)2}$ (1) I_C = 5 A dc I_B = 500 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 f = 1 MHz
	2N5151 (2)	2N5153 (2)	2N5151 (2)	2N5153 (2)			
					V dc	V dc	pF
Min	30	70	6	7			
Max (TO-205)	90	200			2.2	1.5	250
Max (U3)	90	200			2.2	1.5	250

(1) Pulsed see 4.5.1.
(2) The limits specified apply to all package outlines unless otherwise stated.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol 11	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	6
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		7
LD	.016	.021	0.41	0.53	8, 9
LL	See notes 8, 9, 12, 13				
LU	.016	.019	0.041	0.48	8, 9
L_1		.050		1.27	8, 9
L_2	.250		6.35		8, 9
Q		.050		1.27	6
TL	.029	.045	0.74	1.14	4, 5
TW	.028	.034	0.71	0.86	3
r		.010		0.25	11
α	45° TP		45° TP		7
P	.100		2.54		

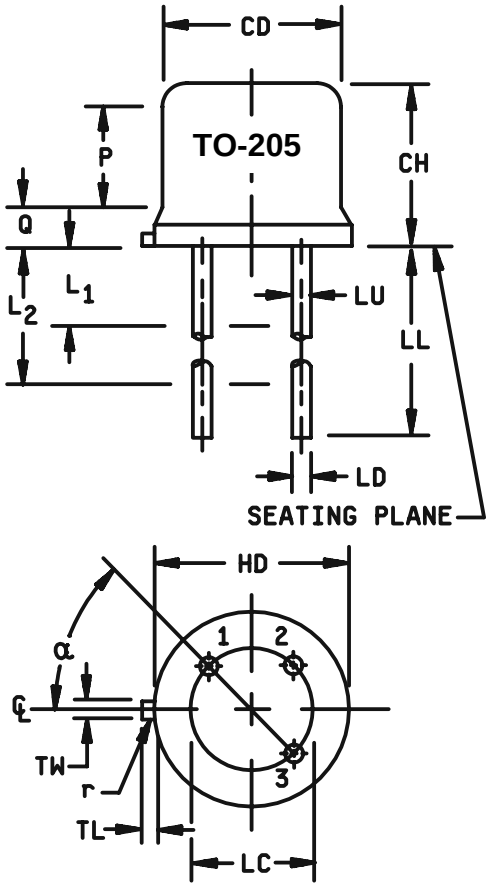


FIGURE 1. Physical dimensions (TO-205).

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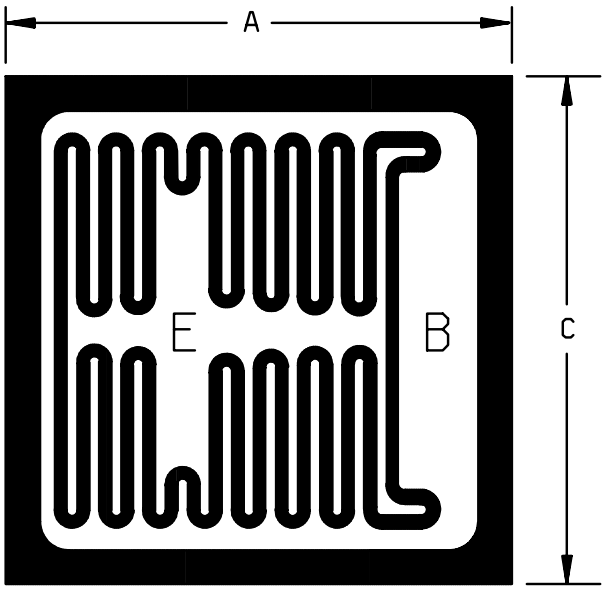
NOTES:

1. Dimensions are in inches.
* 2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 + .001 - .000 (1.37 + 0.03 - 0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
8. LU applied between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. The collector shall be electrically and mechanically connected to the case.
11. r (radius) applies to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
13. For transistor types 2N5151 and 2N5153, LL is .5 (13 mm) minimum, and .75 (19 mm) maximum.
14. For transistor types 2N5151L and 2N5153L, LL is 1.5 (38 mm) minimum and 1.75 (44.4 mm) maximum.
15. Lead designation, depending on device type, shall be as follows:

Lead number	TO-205
1	Emitter
2	Base
3	Collector

FIGURE 1. Physical dimensions (TO-205) - Continued.

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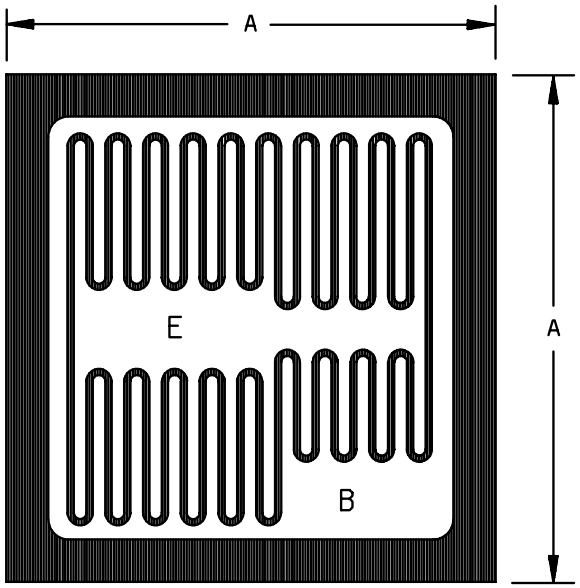


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.117	.127	2.97	3.23
C	.117	.127	2.97	3.23

NOTES:
1. Dimensions are in inches.
* 2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. The physical characteristics of the die are:
   Thickness: .008 (0.20 mm) to .012 (0.30 mm), tolerance is ±.005 (0.13 mm).
   Top metal: Aluminum, 40,000 Å minimum, 50,000 Å nominal.
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = .015 (0.38 mm) x .0072 (0.183 mm).
   E = .015 (0.38 mm) x .0060 (0.152 mm).

FIGURE 2. JANHCA and JANKCA die dimensions.

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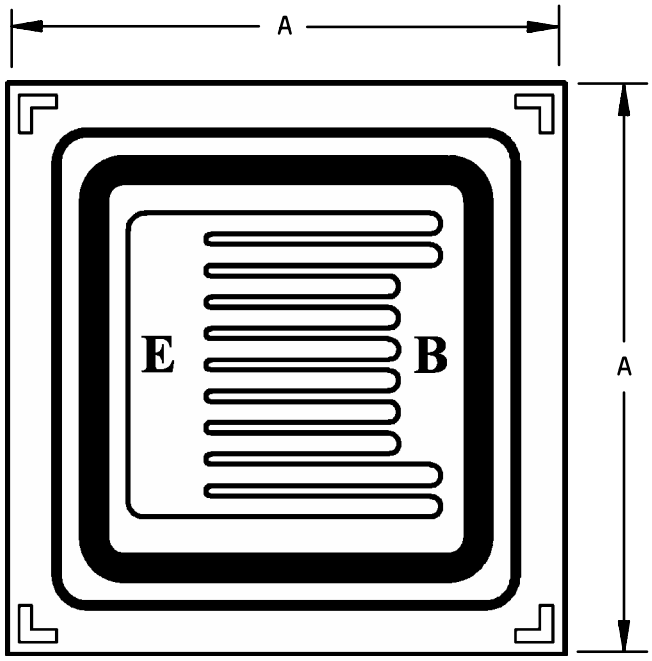


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.100	.105	2.54	2.67

NOTES:
1. Dimensions are in inches.
* 2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The physical characteristics of the die are;
5. Thickness: .0078 (0.198 mm) nominal, tolerance is ±.005 (0.13 mm).
   Top metal: Aluminum, 25,000 Å minimum, 33,000 Å nominal.
   Back metal: Gold 1,500 Å minimum, 2,500 Å nominal.
   Back side: Collector.
   Bonding pad: .012 (0.305 mm) min. x .030 (0.761 mm) minimum.

FIGURE 3. JANHCB and JANKCB die dimensions.

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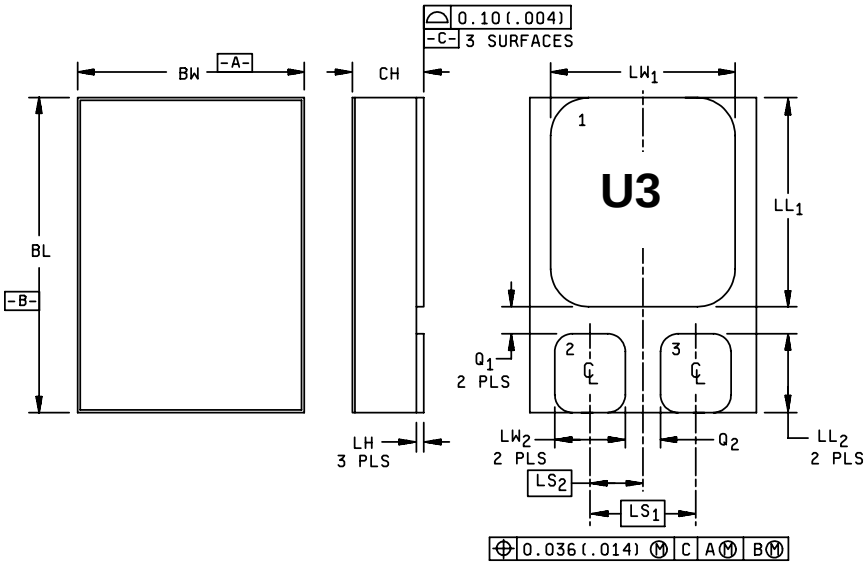


Ltr.	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.126	.130	3.20	3.30

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .010 inches (0.25 mm) ± .0015 inches (0.038 mm) nominal.
   Top metal: Aluminum 30,000Å minimum, 33,000Å nominal.
   Back metal: A. Al/Ti/Ni/Ag15kÅ/2kÅ/7kÅ/7kÅmin.18kÅ/3kÅ/10kÅ/10kÅ nom.
   B. Gold 2,500Å minimum, 3,000Å nominal.
   Back side: Collector.
   Bonding pad: .012 (0.305 mm) min. x .030 (0.761 mm) minimum.

* FIGURE 4. JANHC and JANKC C-version die dimensions.

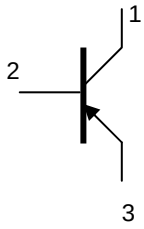
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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.395	.405	10.04	10.28
BW	.291	.301	7.40	7.64
CH	.1085	.1205	2.76	3.06
LH	.010	.020	0.25	0.51
LW1	.281	.291	7.14	7.41
LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84
LL2	.115	.125	2.93	3.17
LS1	.150 BSC		3.81 BSC	
LS2	.075 BSC		1.91 BSC	
Q1	.030		0.762	
Q2	.030		0.762	

NOTES:
1. Dimensions are in inches.
* 2. Millimeters are given for general information only.
3. Terminal 1 - collector, terminal 2 - base, terminal 3 – emitter.

SCHEMATIC



* FIGURE 5. Physical dimensions and configuration for surface mount (U3).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (TO-205), figures 2, 3, and 4 for JANHC and JANKC, figure 5 for U3 herein.

3.4.1 Current density. Current density of internal conductors shall be as specified in MIL-PRF-19500.

3.4.2 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

* 3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500	Measurement	
	JANS levels	JANTX and JANTXV levels
1a 1b	Not required Required	Not required Required for JANTXV only
2	Optional	Optional
3a 3b * (1) 3c	Required Not applicable Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3)	Required Not applicable Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3.).
4	Required	Optional
5	Required	Not applicable
7a and 7b	Optional	Optional
8	Required	Not required
9	I_{CES1} and h_{FE2}	Not applicable
10	48 hours minimum.	48 hours minimum.
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or100 nA dc, whichever is greater. $\Delta h_{FE2} = \pm$ 20 percent.	I_{CES1} and h_{FE2}
12	See 4.3.2	See 4.3.2
13	Subgroup 2 of table I herein; ΔI_{CES1} = 100 percent of initial value or 100 nA dc, whichever is greater. $\Delta h_{FE2} = \pm$ 20 percent.	Subgroup 2 of table I herein; ΔI_{CES1} = 100 percent of initial value or 100 nA dc, whichever is greater. $\Delta h_{FE2} = \pm$ 20 percent.
14a and 14b	Required	Required
15	Required	Not required
16	Required	Not required

(1) Shall be performed anytime after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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* 4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10-30 V dc, T_A = room ambient as defined in the general requirements, 4.5 of MIL-STD-750. Power shall be applied to the device to achieve a junction temperature, T_J = +175 °C minimum and a minimum P_D = 75 percent of PT maximum rated as defined in 1.3 herein.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of Mil-Std-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit used in screen 3c of 4.3 herein and table I shall comply with the thermal impedance graph in figures 10, 11, and 12 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa of MIL-PRF-19500 (JANS) and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein. See 4.4.2.2 herein JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be after each step and shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection table VIa (JANS) of MIL-PRF-19500.

	Subgroup	Method	Conditions
*	B4	1037	V_{CB} = 40 V dc ± 1 V, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
	B5	1027	(NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample). V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3) $T_A \leq$ +35°C. Option 1: 96 hours minimum, sample size in accordance with table VIa of MIL-PRF-19500, adjust P_D to achieve T_J = +275°C minimum. Option 2: 216 hours, sample size = 45, c = 0; adjust P_D to achieve T_J = +225°C minimum.
	B6	3131	See 4.5.2.

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* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new assembly lot option is exercised, the failed assembly lot shall be scrapped.

	Step	Method	Conditions
*	1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
*	2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
*	3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein, and only apply to subgroup C6.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

	Subgroup	Method	Condition
*	C2	2036	Test condition E; (method 2036 not applicable for U3 devices).
*	C6	1026	1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

* 4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

	Subgroup	Method	Condition
*	C2	2036	Test condition E; not applicable for U3 devices.
*	C5	3131	$R_{\theta JA}$ for TO-205 (see 1.3), $R_{\theta JC}$ for U3 (see 1.3).
	C6		Not applicable.

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* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta measurements shall be in accordance with table II herein.

4.5 Methods of inspection and test. Methods of inspection and test shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 500 mA minimum dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be within the range $+25°C \leq T_R \leq +35°C$. The chosen reference temperature shall be recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

* f. See 1.3 for maximum limit of $R_{\theta JC}$.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 1 2/						
	Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
	Solderability 3/ 4/	2026	n = 15 leads, c = 0				
	Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
	Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
	Electrical measurements 4/		Group A, subgroup 2				
*	Hermetic seal 4/ 6/ Fine leak Gross leak	1071	n = 22 devices, c = 0 Test condition G or H Test condition C or D				
	Bond strength 3/ 4/	2037	Precondition: T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
*	De-cap internal visual	2075	n = 4, c = 0				
	Subgroup 2						
*	Thermal impedance 7/	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
	Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 100 mA dc; I_B = 0, pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
	Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 60 V dc; V_{BE} = 0	I_{CES1}		1.0	µA dc
	Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 100 V dc; V_{BE} = 0	I_{CES2}		1.0	mA dc
	Collector to emitter cutoff current	3041	Bias condition D, V_{CE} = 40 V dc; I_B = 0	I_{CEO}		50	µA dc
	Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc; I_C = 0	I_{EBO1}		1.0	µA dc
	Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5.5 V dc; I_C = 0	I_{EBO2}		1.0	mA dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 2 - Continued						
	Forward current transfer ratio 2N5151, L, and U3 2/ 2N5153, L, and U3	3076	V_{CE} = 5 V dc; I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE1}	 20 50		
	Forward current transfer ratio 2N5151, L, and U3 2/ 2N5153, L, and U3	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc, pulsed (see 4.5.1)	h_{FE2}	 30 70	 90 200	
	Forward current transfer ratio 2N5151, L, and U3 2/ 2N5153, L, and U3	3076	V_{CE} = 5 V dc; I_C = 5 A dc, pulsed (see 4.5.1)	h_{FE3}	 20 40		
	Base-emitter voltage (non-saturated)	3066	Test condition B, V_{CE} = 5 V dc; I_C = 2.5 A dc, pulsed (see 4.5.1)	V_{BE}		1.45	V dc
	Base-emitter saturation voltage	3066	Test condition A, I_C = 2.5 A dc; I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.45	V dc
	Base-emitter saturation voltage	3066	Test condition A, I_C = 5 A dc; I_B = 500 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.2	V dc
	Collector-emitter saturation voltage	3071	I_C = 2.5 A dc; I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.75	V dc
	Collector-emitter saturation voltage	3071	I_C = 5 A dc; I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.5	V dc
	Subgroup 3						
	High temperature operation:		T_C = +150°C				
*	Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 60 V dc; V_{BE} = +2 V dc	I_{CEX}		25	µA dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 continued						
Low temperature operation		T_C = -55°C				
Forward - current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc; pulsed (see 4.5.1).	h_{FE4}			
2N5151, L, and U3 2/				15		
2N5153, L, and U3				25		
Subgroup 4						
Common-emitter, small-signal, short-circuit, forward-current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 100 mA dc; f = 1 KHz	h_{fe}			
2N5151, L, and U3 2/				20		
2N5153, L, and U3				50		
Magnitude of common-emitter, small-signal short-circuit, forward-current, transfer ratio	3306	V_{CE} = 5 V dc; I_C = 500 mA dc, f = 10 MHz	$\|h_{fe}\|$			
2N5151, L, and U3 2/				6		
2N5153, L, and U3				7		
Open-circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0, f = 1 MHz	C_{obo}		250	pf
Switching time		I_C = 5 A dc; I_{B1} = 500 mA dc	t_{on}		0.5	µs
		I_{B2} = -500 mA dc	t_s		1.4	µs
		$V_{BE(off)}$ = 3.7 V dc	t_f		0.5	µs
		R_L = 6 Ω, (see figure 14)	t_{off}		1.5	µs

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (dc)	3051	Pre-pulse condition for each test: T_C = +25°C, see figure 15.				
		Pulse condition for each test: t_p = 1 sec. 1 cycle. T_C = +25°C.				
Test # 1		V_{CE} = 5.0 V dc, I_C = 2 A dc for TO-205 V_{CE} = 5.8 V dc, I_C = 2 A dc for U3.				
Test # 2		V_{CE} = 32 V dc, I_C = 310 mA dc for TO-205 V_{CE} = 32 V dc, I_C = 360 mA dc for U3.				
Test # 3		V_{CE} = 80 V dc, I_C = 12.5 mA dc for TO-205 V_{CE} = 80 V dc, I_C = 14.5 mA dc for U3.				
Safe operating area (unclamped inductive)		T_C = +25°C; R_{BB1} = 10 Ω; R_{BB2} = 100 Ω; L = 0.3 mH; RL = 0.1 Ω; V_{CC} = 10 V dc; V_{BB1} = 10 V dc; V_{BB2} = 4 V dc; I_{CM} = 10 A dc (see figure 13)				
End point electrical measurements		See table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Hermetic seal test is an end-point to temperature cycling in addition to electrical measurements.
7/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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* TABLE II. Groups B, C and E delta and electrical measurements. 1/ 2/ 3/ 4/

Steps	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward - current transfer ratio	3076	I_C = 2.5 A dc; V_{CE} = 5 V dc, pulsed (see 4.5.1).	Δh_{FE2}	± 20 percent change from initial reading.		

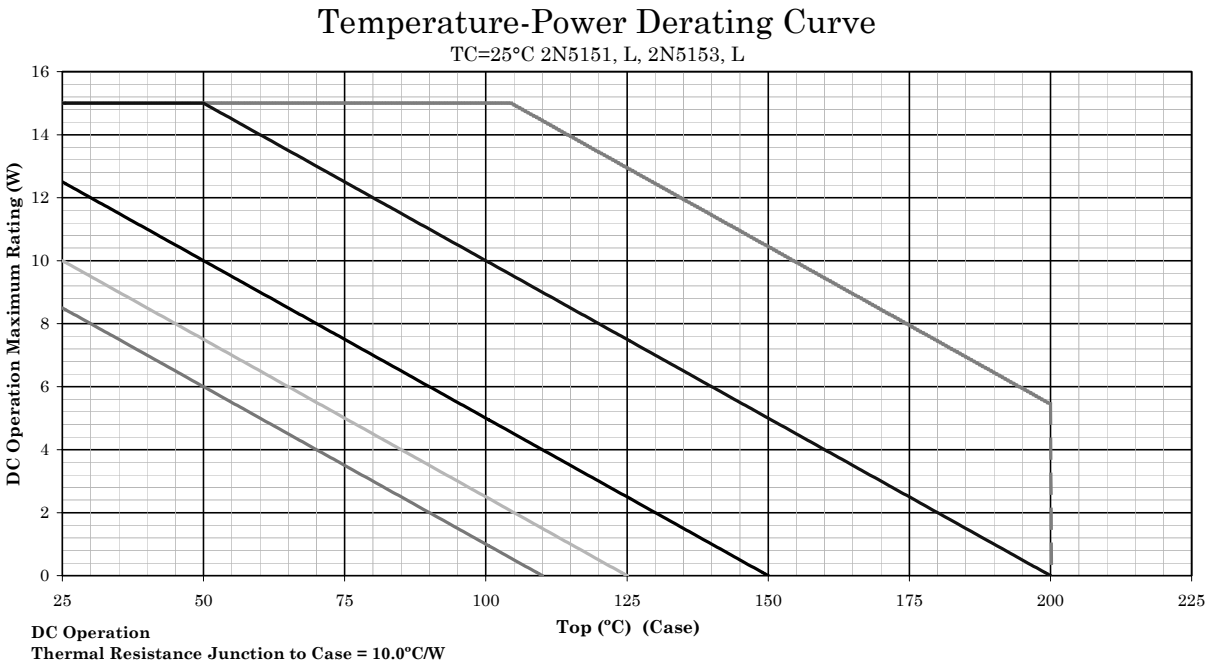
1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows: Subgroups 4 and 5, see table II herein, step 1.
2/ The delta measurements for 4.4.2.2 (JAN, JANTX and JANTXV) for all steps; see table II herein, step 1.
3/ The delta measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 6, see table II herein, step 1.
4/ The delta measurements for table IX of MIL-PRF-19500 and table III herein are as follows: Subgroups 1 and 2, see table II herein, step 1.

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* TABLE III. Group E inspection (all quality levels) – for qualification or re-qualification only.

	Inspection	MIL-STD-750		Qualification
		Method	Conditions	
	Subgroup 1			45 devices c = 0
	Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
*	Hermetic seal Fine leak Gross leak	 1071	 Test conditions G or H Test conditions C or D	
	Electrical measurements		See table I, subgroup 2 and table II herein.	
	Subgroup 2			45 devices c = 0
	Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
	Electrical measurements		See table I, subgroup 2 and table II herein.	
	Subgroup 4			Sample size N/A
*	Thermal impedance curves		Each supplier shall submit their qual-lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	
	Subgroup 5			
	Not applicable			
*	Subgroup 6			3 devices
	ESD	1020		
*	Subgroup 8			45 devices c = 0
	Reverse stability	1033	Condition B	

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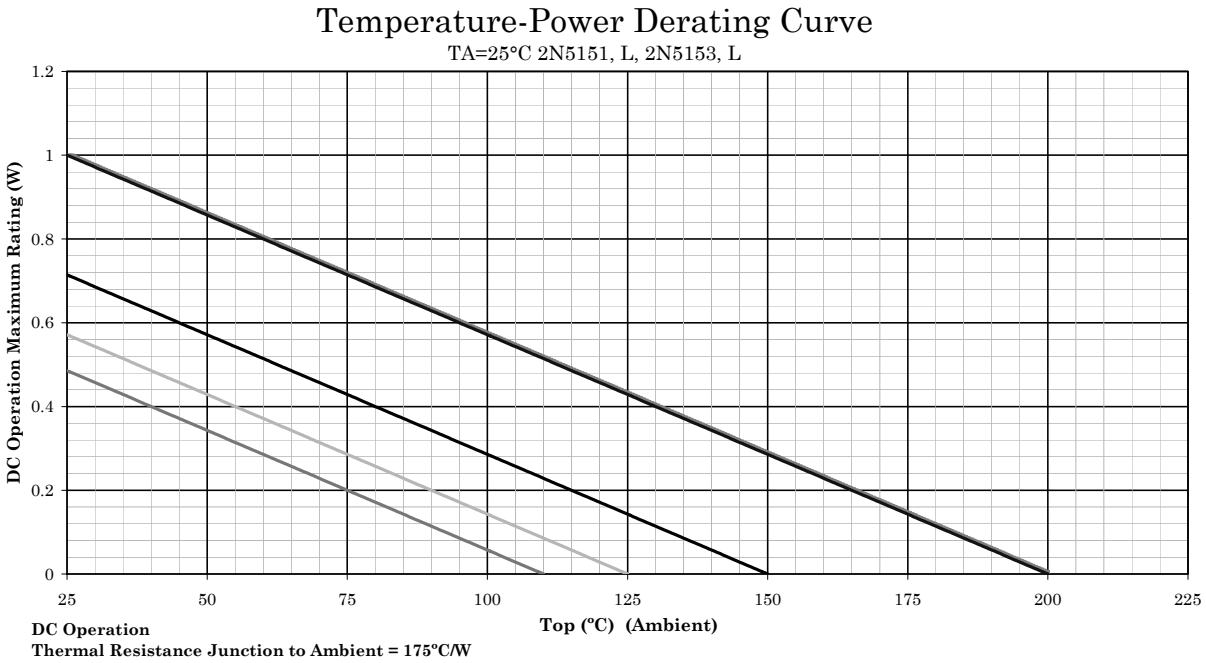


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 6. Temperature-power derating graph, TO-205, case temperature.

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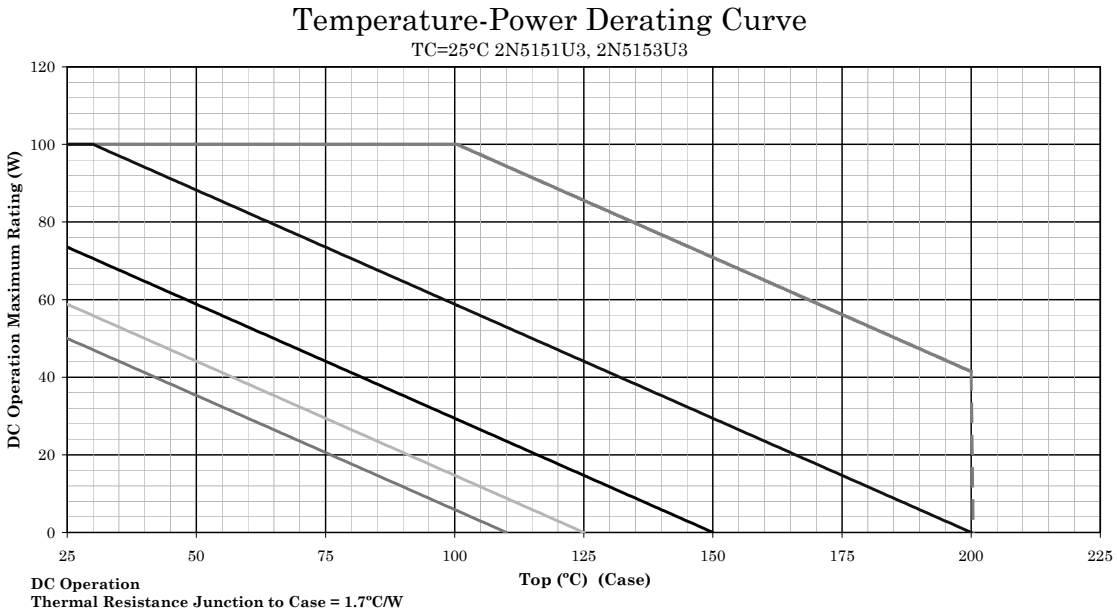


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 7. Temperature-power derating graphs, TO-205 ambient temperature.

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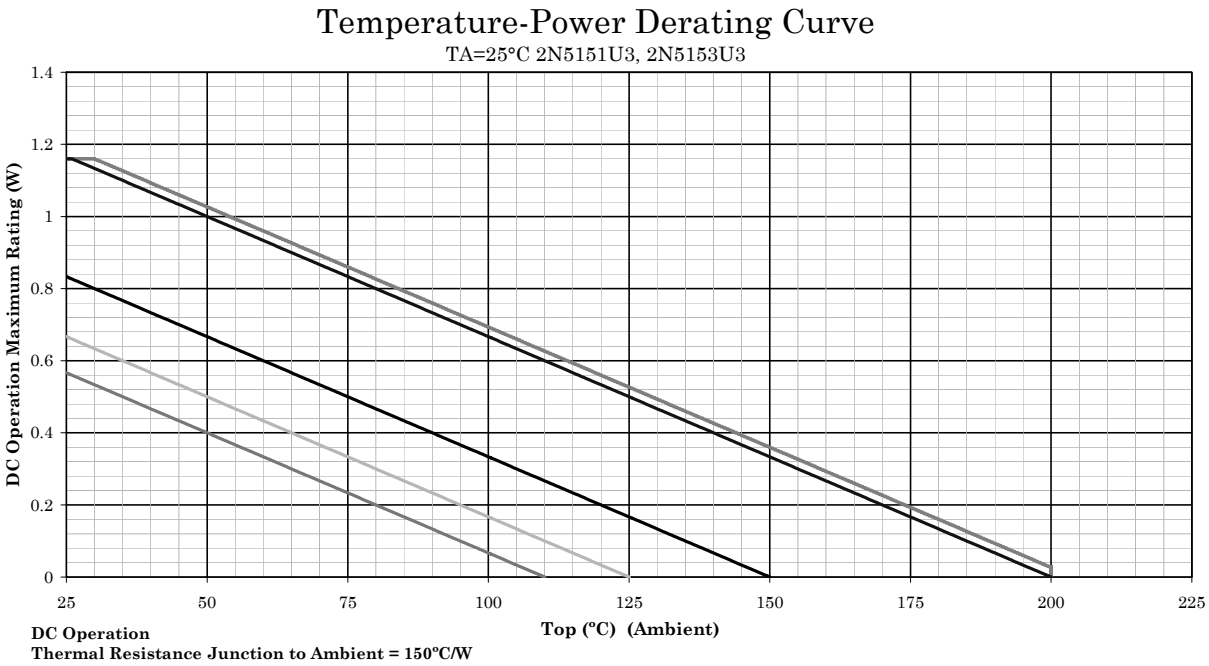


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 8. Temperature-power derating graph, U3 package, case temperature.

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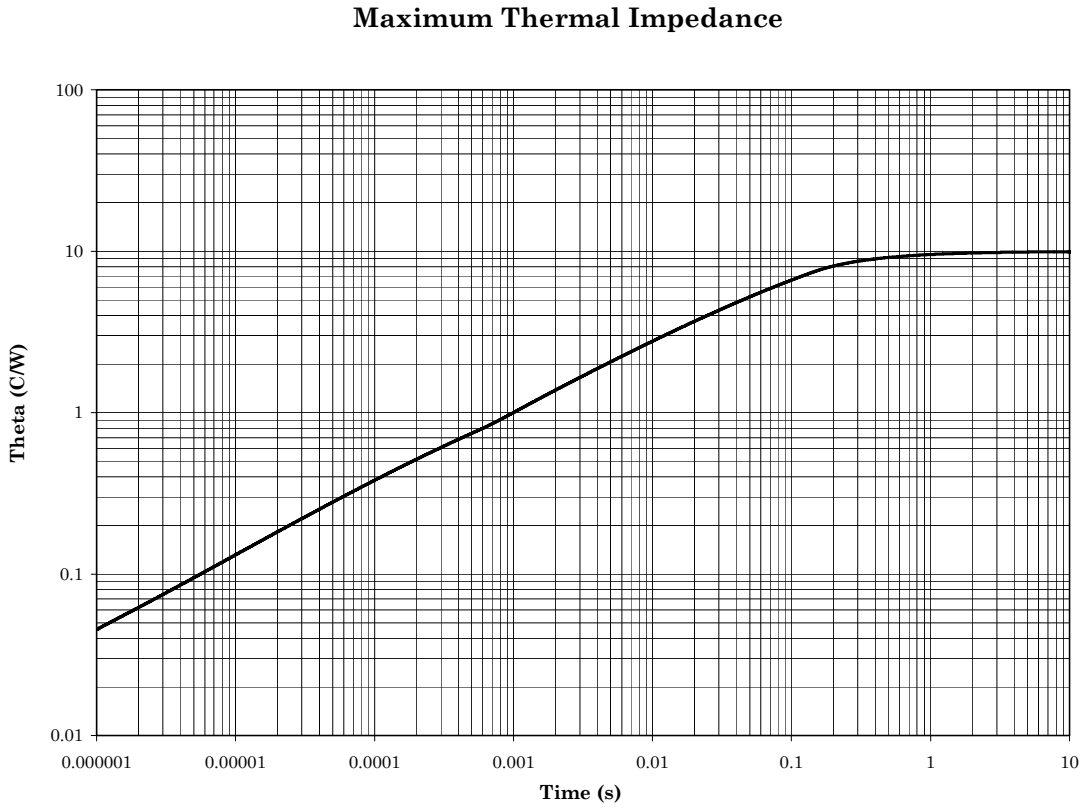


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 9. Temperature-power derating graph, U3 package, ambient temperature.

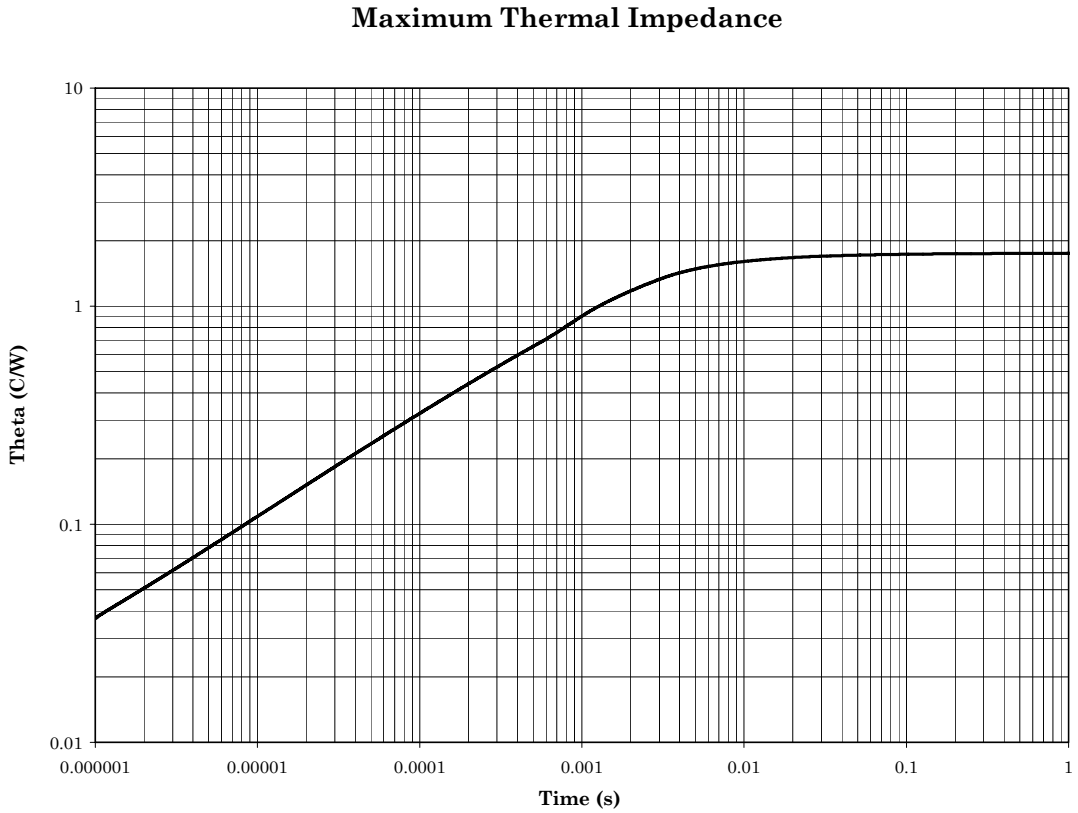
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2N5151, 2N5151L, 2N5153, and 2N5153L at T_C = +25°C, $R_{\theta JC}$ = 10°C/W.

* FIGURE 10. Thermal impedance graph, TO-205 package at case temperature.

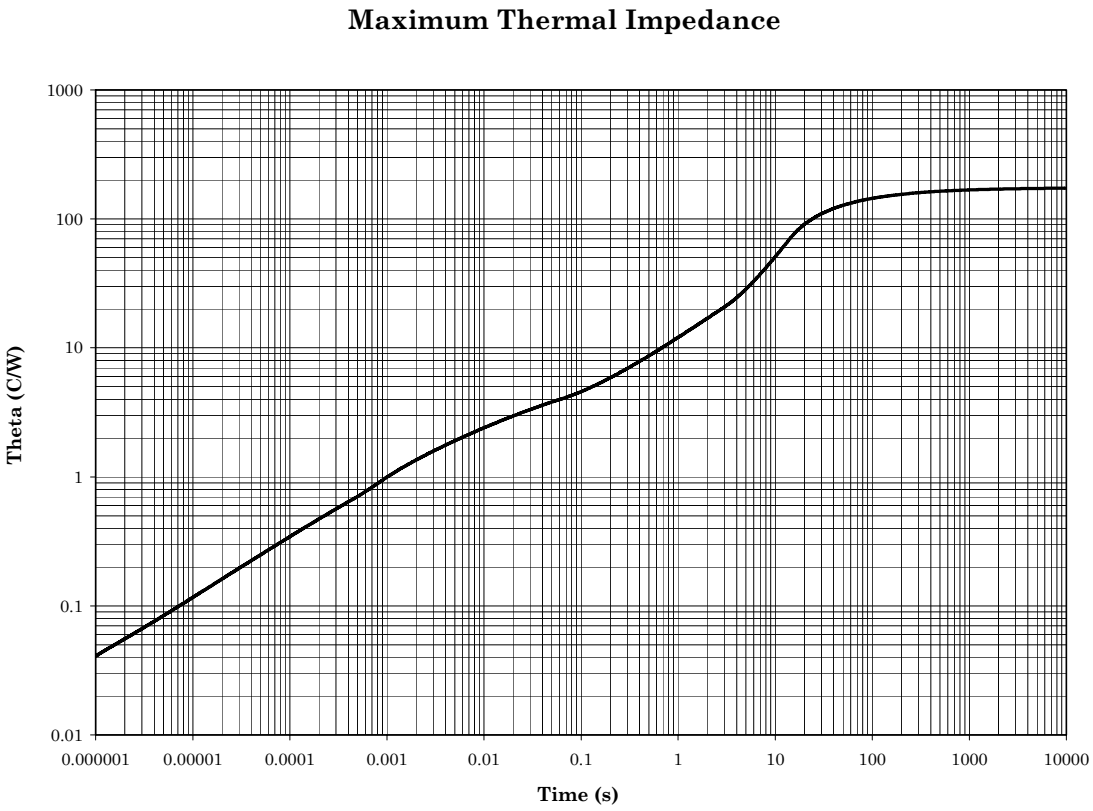
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2N5151U3 and 2N5153U3 at T_C = +25°C, $R_{\theta JC}$ = 1.75°C/W.

* FIGURE 11. Thermal impedance graph, U3 package at case temperature..

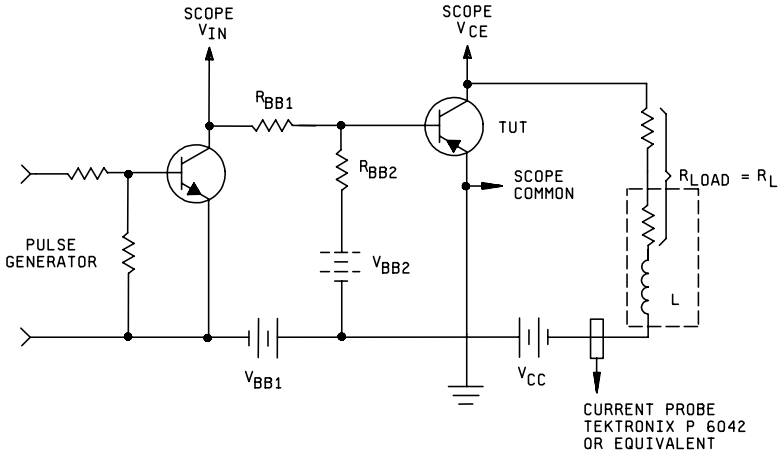
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2N5151, 2N5151L, 2N5153, and 2N5153L at T_A = +25°C, $R_{\theta JA}$ = 175°C/W.
(TO-205)

* FIGURE 12. Thermal impedance graph, TO-205 package at ambient temperature.

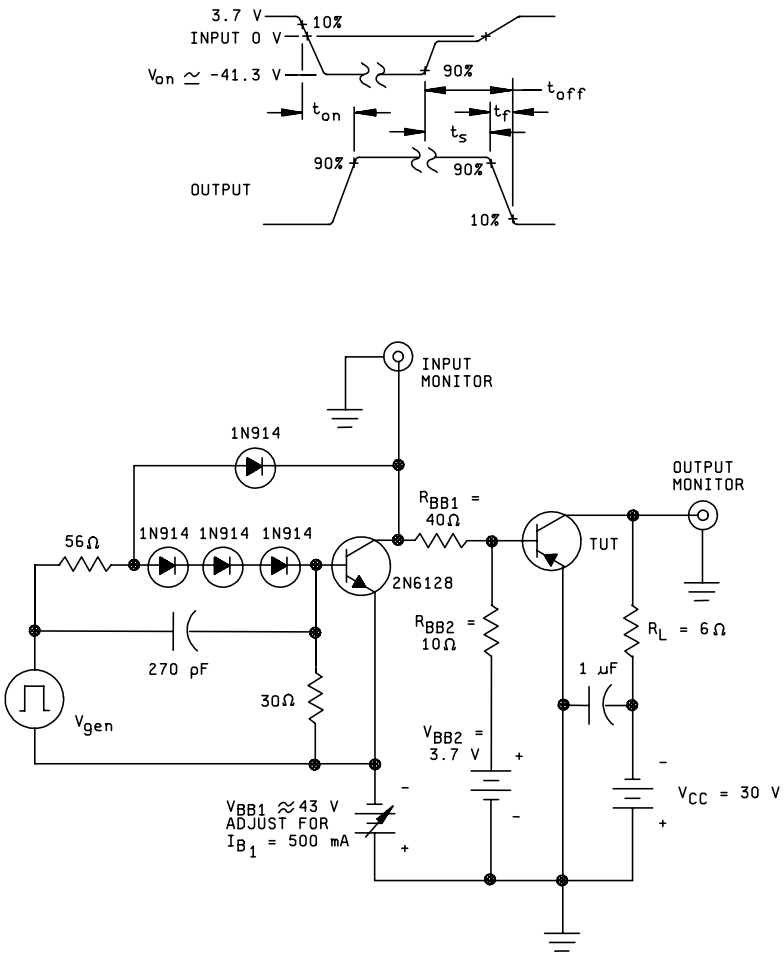
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R_{BB1} = 10 Ω
R_{BB2} = 100 Ω
L = 0.3 mH
R_L = 0.1 Ω
V_{CC} = 10 V dc
I_C = 10 nA
V_{BB1} = 10 V dc
V_{BB2} = 4 V dc

* FIGURE 13. Unclamped inductive load energy test circuit.

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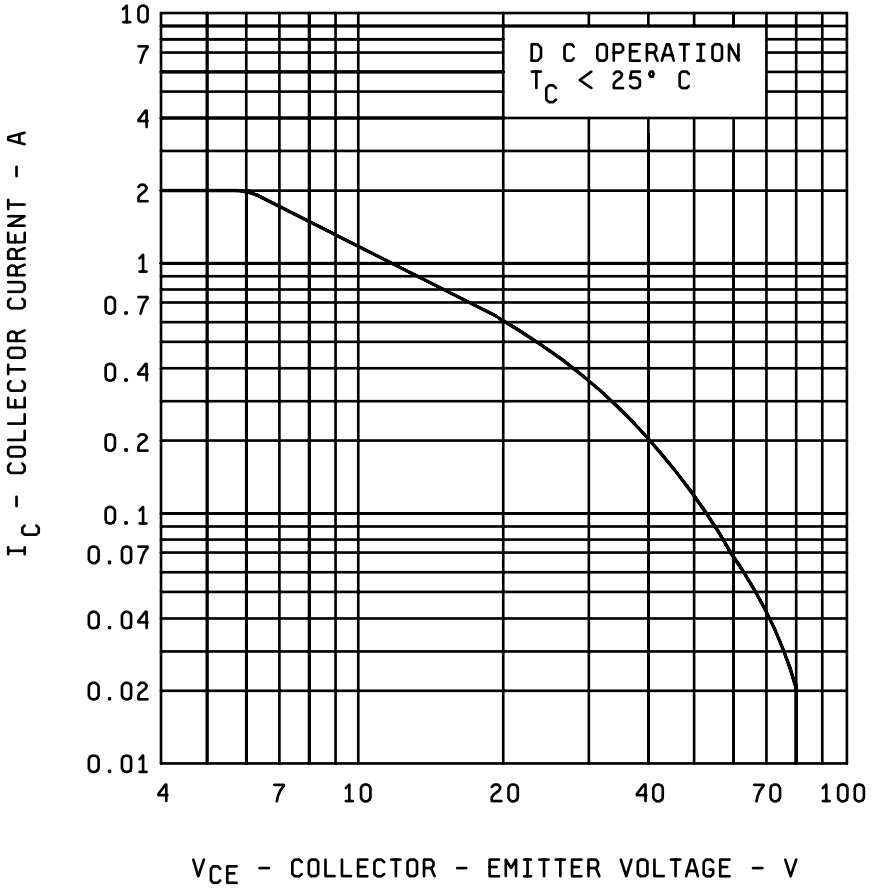


NOTES:

1. V_{gen} is -30 pulse (from 0 V) into a 50 ohm termination.
2. The V_{gen} waveform is supplied by a generator with the following characteristics: $t_r \leq 15$ ns, $t_f = 15$ ns, Z_{OUT} = 50 ohm, duty cycle ≤ 2 percent.
3. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $R_{IN} \geq 10$ MΩ, $C_{IN} \leq 11.5$ pF.
4. Resistors shall be noninductive types.
5. The dc power supplies may require additional bypassing in order to minimize ringing.
6. An equivalent circuit may be used.

* FIGURE 14. Switching time test circuit.

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* FIGURE 15. Maximum safe operating area.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.2).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4. Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCB2N5151) will be identified on the QML.

JANHC and JANKC ordering information			
PIN	Manufacturer		
	33178	34156	43611
2N5151 2N5153	JANHCA2N5151 JANHCA2N5153	JANHCB2N5151 JANHCB2N5153	JANHCC2N5151 JANHCC2N5153
2N5151 2N5153	JANKCA2N5151 JANKCA2N5153	JANKCB2N5151 JANKCB2N5153	JANHCC2N5151 JANHCC2N5153

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
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Review activities:
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